

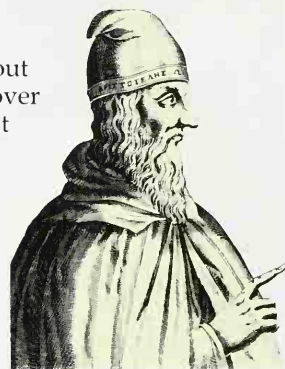


BLADDER CONTROL
When the bladder fills, stretch sensors in its wall tell it to empty. Young children gradually learn to control this automatic reflex action.

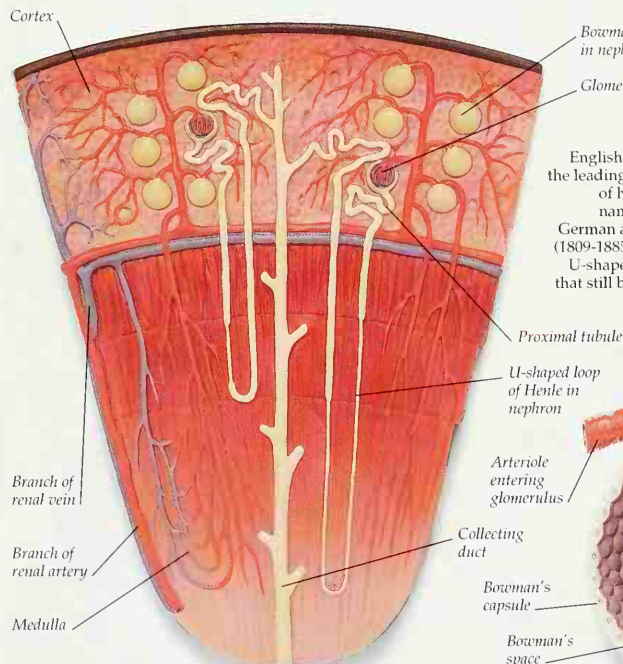
Getting rid of wastes

LIKE ANY COMPLEX MACHINE, the human body produces several waste products. Lungs breathe out carbon dioxide gas; the digestive tract expels leftover food and other matter as feces; the skin sweats out certain salts. Another waste is urine. The kidneys filter undesirable substances and excess water from the blood. Urine is stored in a bag, the bladder, before being expelled. The kidneys and bladder, along with their various connecting

tubes, are known as the urinary, or excretory, system. One of the earliest accurate descriptions of the system was in Aristotle's *Historia Animalium*. It showed the two kidneys, the ureter pipes that connect them to the bladder, and the urethra, the exit tube from the bladder. Study with the naked eye did not reveal how the kidneys worked, so there was little progress from Aristotle until the age of the microscope, when Marcello Malpighi (p.12) identified the glomerulus in the 1650s. Today, we have a detailed understanding of how one million nephrons in each kidney filter blood to make urine.



GIANT OF ANCIENT GREECE
Aristotle (384-322 BC) is known as the "father of nature and biology." Before Aristotle's time, philosophers continued traditional beliefs while paying little attention to the real world around them. Aristotle based his writings on observation, and he was one of the first people to look inside real bodies and record what he saw. He was also a pioneer in the grouping of living things according to features they shared. Such classification underlies modern science.

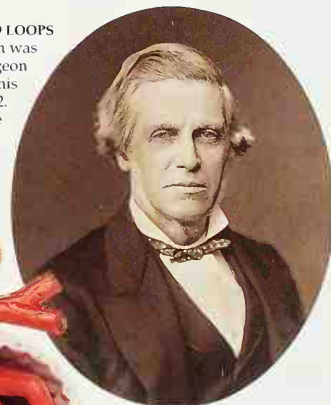


KIDNEY'S NEPHRON

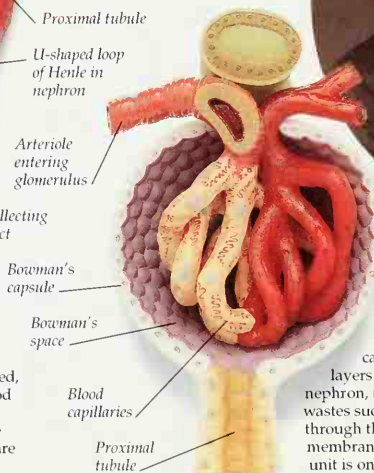
The kidney's filtering units are the nephrons. Above, two are enlarged, passing between the cortex and medulla layers of the kidney. Blood flows through a knot of tiny blood vessels, the glomerulus, in the Bowman's capsule (see right). Salts and minerals pass through the capsule into a U-shaped tube. Here, useful substances and water are reabsorbed into the blood. The rest, urine, flows into a larger duct.

CAPSULES AND LOOPS

Englishman William Bowman was the leading urinary and eye surgeon of his day. He identified his namesake capsule in 1842. German anatomist Jacob Henle (1809-1885) was to describe the U-shaped loop 20 years later that still bears his name today.



William Bowman
(1816-1892)



INSIDE BOWMAN'S CAPSULE

Bowman's microscopic study showed that the tiny knot of capillaries in the glomerulus is surrounded by a two-layered capsule. The space between the layers opens into the next part of the nephron, the proximal tubule. Water and wastes such as urea, salts, and minerals pass through the capillary walls and capsule membrane into the tube. The whole capsule unit is only about 0.008 in (0.2 mm) across.